

HULTSFRED, SwedenWMO: **025650**Lat: **57.526N**Lon: **15.823E**Elev: **112**StdP: **99.99**Time Zone: **1.00 (EUC)**Period: **08-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.0	-12.6	-18.5	0.7	-16.3	-14.1	1.1	-10.6	9.9	5.4	8.8	4.7	0.8	280	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	27.6	17.9	25.7	17.1	23.9	16.2	19.5	24.8	18.5	23.4	17.5	22.0	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.8	13.0	21.0	16.7	12.1	20.0	15.7	11.3	19.2	55.9	25.1	52.8	23.3	49.5	22.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.2	6.3	-19.8	30.3	5.3	2.3	-23.6	32.0	-26.7	33.4	-29.7	34.7	-33.5	36.4		
			-18.8	21.0	4.7	1.0	-22.2	21.7	-25.0	22.3	-27.7	22.9	-31.1	23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	-2.3	-1.2	1.9	5.9	10.6	14.3	16.9	15.6	12.0	7.1	3.5	-0.1	(6)	
	DBStd	7.53	4.82	4.62	4.06	3.46	3.77	3.02	2.79	2.54	2.97	3.45	3.62	5.35	(7)	
	HDD10.0	1743	382	313	253	128	38	3	0	0	14	103	196	314	(8)	
	HDD18.3	4148	641	547	511	372	240	127	62	92	189	349	445	572	(9)	
	CDD10.0	668	0	0	0	6	57	131	215	173	75	12	1	0	(10)	
	CDD18.3	31	0	0	0	0	1	4	19	6	1	0	0	0	(11)	
	CDH23.3	513	0	0	0	0	32	115	268	89	8	0	0	0	(12)	
	CDH26.7	102	0	0	0	0	2	19	60	20	1	0	0	0	(13)	
(14) Wind	WSAvg	2.8	2.8	2.9	3.1	2.9	2.7	2.8	2.6	2.5	2.7	2.7	2.8	3.0	(14)	
(15) Precipitation	PrecAvg	618	37	35	27	33	46	67	88	67	60	53	57	48	(15)	
	PrecMax	758	68	68	47	75	86	138	199	129	154	109	140	87	(16)	
	PrecMin	480	4	7	3	1	8	5	25	7	9	17	11	9	(17)	
	PrecStd	74	18	15	13	21	20	31	41	31	40	28	28	21	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	10.0	15.9	21.4	26.2	29.2	30.6	29.1	24.6	18.4	12.8	10.7	(19)	
		MCWB	7.0	5.5	9.2	12.0	16.4	17.8	19.1	18.7	17.0	13.7	11.2	9.1	(20)	
	2%	DB	6.2	7.7	12.8	18.5	24.0	26.4	28.3	25.6	21.3	15.3	10.9	9.2	(21)	
		MCWB	5.1	5.2	7.2	10.5	14.8	16.6	18.2	18.2	15.7	12.7	9.6	7.9	(22)	
	5%	DB	5.0	6.0	10.1	15.9	21.9	24.2	26.3	23.5	19.4	13.8	9.6	7.5	(23)	
		MCWB	3.9	4.3	6.1	9.3	13.8	15.7	17.8	16.7	14.5	11.6	8.5	6.2	(24)	
	10%	DB	3.7	4.7	8.0	13.6	19.3	22.1	24.3	21.9	17.6	12.5	8.4	6.0	(25)	
		MCWB	2.8	3.3	5.0	8.2	12.7	14.5	16.8	15.9	13.6	10.6	7.5	4.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.0	6.9	9.8	13.0	17.5	19.9	21.0	20.6	18.3	14.7	11.4	9.3	(27)	
		MCDB	8.0	8.6	14.4	19.1	23.8	25.5	26.2	25.3	22.5	16.9	12.3	10.4	(28)	
	2%	WB	5.3	5.5	8.2	11.5	16.1	18.2	19.7	19.1	16.5	13.1	9.7	8.0	(29)	
		MCDB	6.1	7.1	12.6	17.1	21.9	23.9	25.4	24.0	19.7	14.9	10.6	9.2	(30)	
	5%	WB	4.1	4.4	6.6	10.1	14.7	16.5	18.8	17.9	15.4	12.0	8.6	6.4	(31)	
		MCDB	4.9	5.9	9.4	14.9	19.6	22.4	24.1	21.7	18.2	13.7	9.4	7.3	(32)	
	10%	WB	2.8	3.4	5.4	8.8	13.2	15.2	17.9	16.9	14.4	10.9	7.6	5.1	(33)	
		MCDB	3.6	4.7	7.8	13.2	18.2	20.6	22.5	20.6	16.7	12.3	8.3	5.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.5	6.3	9.0	12.1	12.9	12.4	11.8	11.6	10.0	7.7	5.1	5.3	(35)	
		MCDBR	5.6	7.9	12.4	17.2	17.5	17.0	15.7	14.3	12.9	9.2	6.3	5.6	(36)	
	5% WB	MCWBR	5.0	5.8	7.7	9.9	9.0	8.1	6.9	7.4	7.6	6.6	5.5	5.2	(37)	
		MCWBR	5.4	7.0	11.2	14.7	14.3	14.8	12.8	12.1	10.9	8.6	5.7	5.6	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.289	0.321	0.338	0.375	0.373	0.362	0.389	0.382	0.351	0.341	0.316	0.277	(40)	
		taud	2.283	2.324	2.360	2.327	2.394	2.448	2.392	2.407	2.482	2.429	2.328	2.227	(41)	
	Edn at Noon	Edn at Noon	578	721	814	835	863	877	842	821	797	693	537	471	(42)	
		Edh at Noon	48	71	89	108	107	103	107	99	79	64	47	38	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.37	0.97	2.47	4.03	5.26	5.70	5.16	4.17	2.83	1.29	0.48	0.26	(44)	
	RadStd	RadStd	0.06	0.19	0.26	0.51	0.47	0.29	0.60	0.41	0.37	0.22	0.09	0.03	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (48 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-125	-162	N/A	N/A

Nomenclature: See separate page